POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5043B | Semester V | Theory | 4 Credits

Microwave Devices & Radar

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5043B as Microwave Devices & Radar in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Microwave Engineering course and polytechnic microwave curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. RF designs, radar calibrations, equipment specifications and power levels must be based on approved engineering plans, certified hardware data, current regulatory standards and competent professional review.

Verified source note: Verified sources support waveguides, microwave tubes (Klystron, Magnetron, TWT), semiconductor devices (Gunn, IMPATT), and radar fundamentals. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Microwave Devices & Radar studies the generation, transmission and detection of electromagnetic waves in the gigahertz frequency range. It develops the ability to analyze waveguides and microwave components, evaluate microwave tubes (Klystron, Magnetron) and semiconductor devices (Gunn, IMPATT), and understand the principles of radar range and MTI systems while respecting the technical and regulatory boundaries of RF and strategic communication engineering.

> **Malayalam support:** ▯ handbook ▯▯▯▯▯▯▯▯▯▯▯▯▯ syllabus topic ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯▯▯▯▯▯ principle, diagram/procedure, application, common mistake ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯.

Prerequisite foundation

Electromagnetic wave theory, analog communication, digital electronics, antenna engineering and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain microwave fundamentals, transmission modes and the characteristics of rectangular and circular waveguides.
2. Explain the working principles and applications of passive microwave components like isolators and circulators.
3. Explain the operation of microwave tubes and semiconductor devices for high-frequency signal generation.
4. Explain radar principles, the radar range equation and the operation of pulse, Doppler and MTI radar systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes		
CO	Official outcome	Bloom level
CO1	Explain microwave transmission modes and the properties of waveguides and resonators.	Understand
CO2	Explain the working of passive microwave components and scattering parameters.	Understand
CO3	Explain the operation of microwave tubes and solid-state microwave devices.	Understand
CO4	Explain radar system principles, range analysis and specialized radar techniques.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Waveguides and Resonators	Microwave spectrum and advantages; Rectangular waveguides (TE and TM modes); cutoff frequency; phase and group velocity; Circular waveguides; Waveguide resonators; Smith chart basics.	Approx. 14 hours	CO1	Module 1
2	Microwave Components and S-Parameters	Scattering matrix (S-parameters); Microwave T-junctions (E-plane, H-plane, Magic Tee); Directional couplers; Isolators and Circulators (Ferrite devices); Attenuators and Phase shifters.	Approx. 14 hours	CO2	Module 2
3	Microwave Tubes and Semiconductor Devices	Limitations of conventional tubes; Klystron (Two-cavity and Reflex); Magnetron; Traveling Wave Tube (TWT); Microwave semiconductor devices: Gunn diode, IMPATT, TRAPATT, PIN diode.	Approx. 16 hours	CO3	Module 3
4	Radar Fundamentals and Systems	Radar block diagram; Radar range equation; Pulse radar; Continuous Wave (CW) radar; Doppler effect; Moving Target Indicator (MTI) radar; Delay line cancellers; Radar displays (A-scope, PPI).	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Waveguides and Resonators

Microwave spectrum and advantages; Rectangular waveguides (TE and TM modes); cutoff frequency; phase and group velocity; Circular waveguides; Waveguide resonators; Smith chart basics.

CO1 · Approx. 14 hours

Microwave Components and S-Parameters

Scattering matrix (S-parameters); Microwave T-junctions (E-plane, H-plane, Magic Tee); Directional couplers; Isolators and Circulators (Ferrite devices); Attenuators and Phase shifters.

CO2 · Approx. 14 hours

Microwave Tubes and Semiconductor Devices

Limitations of conventional tubes; Klystron (Two-cavity and Reflex); Magnetron; Traveling Wave Tube (TWT); Microwave semiconductor devices: Gunn diode, IMPATT, TRAPATT, PIN diode.

CO3 · Approx. 16 hours

Radar Fundamentals and Systems

Radar block diagram; Radar range equation; Pulse radar; Continuous Wave (CW) radar; Doppler effect; Moving Target Indicator (MTI) radar; Delay line cancellers; Radar displays (A-scope, PPI).

CO4 · Approx. 16 hours

MODULE 1

Waveguides and Resonators

Microwave spectrum and advantages; Rectangular waveguides (TE and TM modes); cutoff frequency; phase and group velocity; Circular waveguides; Waveguide resonators; Smith chart basics.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

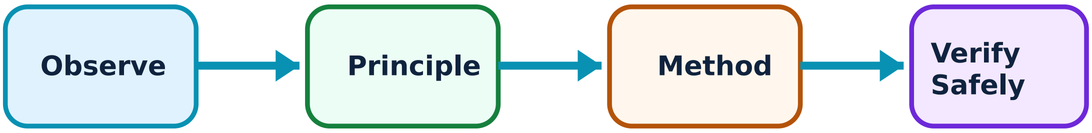
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Waveguides and Resonators: identify the system, state its principle, follow the correct method and verify the result safely.

1. Waveguide principles and modes−

Waveguides are metallic pipes used to transmit microwave energy with low loss. Rectangular waveguides support TE (Transverse Electric) and TM (Transverse Magnetic) modes. The dominant mode (TE10) has the lowest cutoff frequency. Understanding cutoff frequency and wave velocity is essential for designing efficient microwave links.

Study and application method

Calculate the cutoff frequency conceptually for a TE10 mode in a rectangular waveguide; explain the difference between 'Phase Velocity' and 'Group Velocity'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the cutoff frequency conceptually for a TE10 mode in a rectangular waveguide; explain the difference between 'Phase Velocity' and 'Group Velocity'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Waveguides have sharp edges and can carry high power. Do not handle energized waveguides without authorised power-off verification and protective gloves.

2. Resonators and Smith chart−

Waveguide resonators are used as filters and frequency meters. The Smith chart is a graphical tool used to analyze transmission lines and match impedances, converting complex reflection coefficients into normalized impedances.

Study and application method

Describe the working of a rectangular cavity resonator; explain how the Smith chart is used for impedance matching conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of a rectangular cavity resonator; explain how the Smith chart is used for impedance matching conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Microwave resonators are highly sensitive to dimensions. Do not modify resonator cavities without authorised precision machining and frequency calibration.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Microwave Components and S-Parameters

Scattering matrix (S-parameters); Microwave T-junctions (E-plane, H-plane, Magic Tee); Directional couplers; Isolators and Circulators (Ferrite devices); Attenuators and Phase shifters.

Approx. 14 hours | CO2 | Understand · Apply

Learning goals

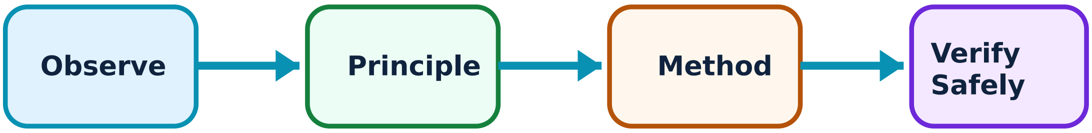
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Microwave Components and S-Parameters: identify the system, state its principle, follow the correct method and verify the result safely.

1. Passive microwave components−

Passive components manage the flow of microwave signals. Isolators allow power flow in one direction only, protecting sources from reflections. Magic Tees are used for mixing and power splitting. Directional couplers sample power from a transmission line for monitoring.

Study and application method

Sketch a 'Magic Tee' and explain its applications in microwave systems; describe the role of a 'Circulator' in a radar duplexer.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a 'Magic Tee' and explain its applications in microwave systems; describe the role of a 'Circulator' in a radar duplexer.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Ferrite devices (isolators/circulators) are sensitive to external magnetic fields. Keep devices away from authorised magnets and magnetic materials to prevent performance degradation.

2. Scattering parameters (S-parameters)−

At microwave frequencies, traditional voltage and current measurements are difficult. S-parameters describe the relationship between incident and reflected waves at the ports of a microwave network, essential for characterizing multi-port devices like couplers and filters.

Study and application method

Define the 'Scattering Matrix' for a two-port network; explain why S-parameters are preferred over ABCD parameters at microwave frequencies.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define the 'Scattering Matrix' for a two-port network; explain why S-parameters are preferred over ABCD parameters at microwave frequencies.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: S-parameter measurements require calibrated Vector Network Analyzers (VNA). Do not perform measurements without authorised VNA calibration and high-quality RF cables.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Microwave Tubes and Semiconductor Devices

Limitations of conventional tubes; Klystron (Two-cavity and Reflex); Magnetron; Traveling Wave Tube (TWT); Microwave semiconductor devices: Gunn diode, IMPATT, TRAPATT, PIN diode.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

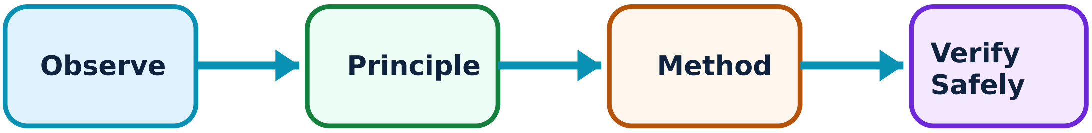
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Microwave Tubes and Semiconductor Devices: identify the system, state its principle, follow the correct method and verify the result safely.

1. Microwave vacuum tubes−

Conventional tubes fail at microwaves due to transit time effects. Microwave tubes use velocity modulation. Klystrons are used as amplifiers or oscillators (Reflex). Magnetrons generate high-power pulses for radar. TWTs provide wideband amplification for satellite communication.

Study and application method

Describe the working principle of a 'Reflex Klystron' oscillator; compare the applications of a Magnetron and a TWT.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working principle of a 'Reflex Klystron' oscillator; compare the applications of a Magnetron and a TWT.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Microwave tubes involve high voltages and produce X-rays. Do not operate tubes without authorised shielding and high-voltage safety interlocks.

2. Solid-state microwave devices−

Semiconductor devices like Gunn diodes (Transferred Electron Effect) and IMPATT diodes (Impact Ionization) generate microwave power. PIN diodes are used as high-speed RF switches and attenuators. These devices enable compact and reliable microwave systems.

Study and application method

Explain the 'Gunn Effect' and its use in microwave oscillators; describe the operation of a PIN diode as an RF switch.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Gunn Effect' and its use in microwave oscillators; describe the operation of a PIN diode as an RF switch.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□□ result □□□□□□□□□□□□ application □□□□□□□□□ □□□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Microwave semiconductor devices are sensitive to ESD (Electrostatic Discharge) and over-voltage. Follow authorised anti-static procedures during handling and installation.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Radar Fundamentals and Systems

Radar block diagram; Radar range equation; Pulse radar; Continuous Wave (CW) radar; Doppler effect; Moving Target Indicator (MTI) radar; Delay line cancellers; Radar displays (A-scope, PPI).

Approx. 16 hours **CO4** **Understand · Apply**

Learning goals

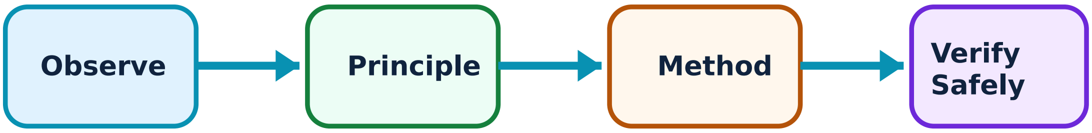
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Radar Fundamentals and Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Radar principles and range−

Radar (Radio Detection and Ranging) uses reflected EM waves to detect targets. The radar range equation relates transmit power, antenna gain, target cross-section and receiver sensitivity to the maximum detection distance. Pulse radar measures range using time delay.

Study and application method

Derive the basic 'Radar Range Equation' conceptually; explain how range is determined in a pulse radar system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Derive the basic 'Radar Range Equation' conceptually; explain how range is determined in a pulse radar system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Radar systems emit high-power RF pulses. Do not stand in front of an active radar antenna to avoid authorised RF radiation exposure hazards.

2. Doppler and MTI radar−

CW radar uses the Doppler effect to measure target velocity. MTI (Moving Target Indicator) radar uses delay line cancellers to filter out stationary 'clutter' (e.g., buildings, hills) and display only moving targets, essential for air traffic control and defense.

Study and application method

Describe the working of an MTI radar using a block diagram; explain the concept of 'Blind Speeds' in MTI radar.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of an MTI radar using a block diagram; explain the concept of 'Blind Speeds' in MTI radar.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Radar data interpretation requires trained personnel. Do not use radar displays for navigation or tactical decisions without authorised training and system verification.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Waveguides and Resonators

Use the concept flow to revise observation, principle, method and verification.

Module 2: Microwave Components and S-Parameters

Use the concept flow to revise observation, principle, method and verification.

Module 3: Microwave Tubes and Semiconductor Devices

Use the concept flow to revise observation, principle, method and verification.

Module 4: Radar Fundamentals and Systems

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Klystron, Magnetron and TWT in a conceptual table showing their gain and power levels.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the TE10 mode field distribution in a rectangular waveguide.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a basic pulse radar system and label the duplexer.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the maximum radar range conceptually given the peak power and target cross-section.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the type of microwave antenna (e.g., dish, horn) used in a satellite TV receiver or a speed radar.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Waveguides and Resonators

Explain Waveguide principles and modes and state one correct method, application or precaution.−

Waveguides are metallic pipes used to transmit microwave energy with low loss. Rectangular waveguides support TE (Transverse Electric) and TM (Transverse Magnetic) modes. The dominant mode (TE10) has the lowest cutoff frequency. Understanding cutoff frequency and wave velocity is essential for designing efficient microwave links.

Answer framework: Calculate the cutoff frequency conceptually for a TE10 mode in a rectangular waveguide; explain the difference between 'Phase Velocity' and 'Group Velocity'.

Important point: Waveguides have sharp edges and can carry high power. Do not handle energized waveguides without authorised power-off verification and protective gloves.

Explain Resonators and Smith chart and state one correct method, application or precaution.−

Waveguide resonators are used as filters and frequency meters. The Smith chart is a graphical tool used to analyze transmission lines and match impedances, converting complex reflection coefficients into normalized impedances.

Answer framework: Describe the working of a rectangular cavity resonator; explain how the Smith chart is used for impedance matching conceptually.

Important point: Microwave resonators are highly sensitive to dimensions. Do not modify resonator cavities without authorised precision machining and frequency calibration.

Module 2 — Microwave Components and S-Parameters

Explain Passive microwave components and state one correct method, application or precaution.−

Passive components manage the flow of microwave signals. Isolators allow power flow in one direction only, protecting sources from reflections. Magic Tees are used for mixing and power splitting. Directional couplers sample power from a transmission line for monitoring.

Answer framework: Sketch a 'Magic Tee' and explain its applications in microwave systems; describe the role of a 'Circulator' in a radar duplexer.

Important point: Ferrite devices (isolators/circulators) are sensitive to external magnetic fields. Keep devices away from authorised magnets and magnetic materials to prevent performance degradation.

Explain Scattering parameters (S-parameters) and state one correct method, application or precaution.−

At microwave frequencies, traditional voltage and current measurements are difficult. S-parameters describe the relationship between incident and reflected waves at the ports of a microwave network, essential for characterizing multi-port devices like couplers and filters.

Answer framework: Define the 'Scattering Matrix' for a two-port network; explain why S-parameters are preferred over ABCD parameters at microwave frequencies.

Important point: S-parameter measurements require calibrated Vector Network Analyzers (VNA). Do not perform measurements without authorised VNA calibration and high-quality RF cables.

Module 3 — Microwave Tubes and Semiconductor Devices

Explain Microwave vacuum tubes and state one correct method, application or precaution.−

Conventional tubes fail at microwaves due to transit time effects. Microwave tubes use velocity modulation. Klystrons are used as amplifiers or oscillators (Reflex). Magnetrons generate high-power pulses for radar. TWTs provide wideband amplification for satellite communication.

Answer framework: Describe the working principle of a 'Reflex Klystron' oscillator; compare the applications of a Magnetron and a TWT.

Important point: Microwave tubes involve high voltages and produce X-rays. Do not operate tubes without authorised shielding and high-voltage safety interlocks.

Explain Solid-state microwave devices and state one correct method, application or precaution.−

Semiconductor devices like Gunn diodes (Transferred Electron Effect) and IMPATT diodes (Impact Ionization) generate microwave power. PIN diodes are used as high-speed RF switches and attenuators. These devices enable compact and reliable microwave systems.

Answer framework: Explain the 'Gunn Effect' and its use in microwave oscillators; describe the operation of a PIN diode as an RF switch.

Important point: Microwave semiconductor devices are sensitive to ESD (Electrostatic Discharge) and over-voltage. Follow authorised anti-static procedures during handling and installation.

Module 4 — Radar Fundamentals and Systems

Explain Radar principles and range and state one correct method, application or precaution.−

Radar (Radio Detection and Ranging) uses reflected EM waves to detect targets. The radar range equation relates transmit power, antenna gain, target cross-section and receiver sensitivity to the maximum detection distance. Pulse radar measures range using time delay.

Answer framework: Derive the basic 'Radar Range Equation' conceptually; explain how range is determined in a pulse radar system.

Important point: Radar systems emit high-power RF pulses. Do not stand in front of an active radar antenna to avoid authorised RF radiation exposure hazards.

Explain Doppler and MTI radar and state one correct method, application or precaution.−

CW radar uses the Doppler effect to measure target velocity. MTI (Moving Target Indicator) radar uses delay line cancellers to filter out stationary 'clutter' (e.g., buildings, hills) and display only moving targets, essential for air traffic control and defense.

Answer framework: Describe the working of an MTI radar using a block diagram; explain the concept of 'Blind Speeds' in MTI radar.

Important point: Radar data interpretation requires trained personnel. Do not use radar displays for navigation or tactical decisions without authorised training and system verification.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Waveguides and Resonators.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Microwave Components and S-Parameters.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Microwave Tubes and Semiconductor Devices.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Radar Fundamentals and Systems.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Microwave spectrum and advantages; Rectangular waveguides (TE and TM modes); cutoff frequency; phase and group velocity; Circular waveguides; Waveguide resonators; Smith chart basics..
2. Develop a complete answer or practical plan for: Scattering matrix (S-parameters); Microwave T-junctions (E-plane, H-plane, Magic Tee); Directional couplers; Isolators and Circulators (Ferrite devices); Attenuators and Phase shifters..
3. Develop a complete answer or practical plan for: Limitations of conventional tubes; Klystron (Two-cavity and Reflex); Magnetron; Traveling Wave Tube (TWT); Microwave semiconductor devices: Gunn diode, IMPATT, TRAPATT, PIN diode..
4. Develop a complete answer or practical plan for: Radar block diagram; Radar range equation; Pulse radar; Continuous Wave (CW) radar; Doppler effect; Moving Target Indicator (MTI) radar; Delay line cancellers; Radar displays (A-scope, PPI)..

LAST-MILE REVISION

Quick Revision

Module 1: Waveguides and Resonators

Microwave spectrum and advantages; Rectangular waveguides (TE and TM modes); cutoff frequency; phase and group velocity; Circular waveguides; Waveguide resonators; Smith chart basics.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Microwave Components and S-Parameters

Scattering matrix (S-parameters); Microwave T-junctions (E-plane, H-plane, Magic Tee); Directional couplers; Isolators and Circulators (Ferrite devices); Attenuators and Phase shifters.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Microwave Tubes and Semiconductor Devices

Limitations of conventional tubes; Klystron (Two-cavity and Reflex); Magnetron; Traveling Wave Tube (TWT); Microwave semiconductor devices: Gunn diode, IMPATT, TRAPATT, PIN diode.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Radar Fundamentals and Systems

Radar block diagram; Radar range equation; Pulse radar; Continuous Wave (CW) radar; Doppler effect; Moving Target Indicator (MTI) radar; Delay line cancellers; Radar displays (A-scope, PPI).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Waveguide principles and modes	Waveguides are metallic pipes used to transmit microwave energy with low loss.
Resonators and Smith chart	Waveguide resonators are used as filters and frequency meters.
Passive microwave components	Passive components manage the flow of microwave signals.
Scattering parameters (S-parameters)	At microwave frequencies, traditional voltage and current measurements are difficult.
Microwave vacuum tubes	Conventional tubes fail at microwaves due to transit time effects.
Solid-state microwave devices	Semiconductor devices like Gunn diodes (Transferred Electron Effect) and IMPATT diodes (Impact Ionization) generate microwave power.
Radar principles and range	Radar (Radio Detection and Ranging) uses reflected EM waves to detect targets.
Doppler and MTI radar	CW radar uses the Doppler effect to measure target velocity.

LEARNING RESOURCES

References

Recommended microwave-radar references

1. Samuel Y. Liao, Microwave Devices and Circuits.
2. M. Kulkarni, Microwave and Radar Engineering.
3. Merrill I. Skolnik, Introduction to Radar Systems.
4. Annapurna Das and Sisir K. Das, Microwave Engineering.

Approved public microwave-radar sources

- https://nptel.ac.in/courses/108103141
- https://onlinecourses.nptel.ac.in/noc24_ee69/preview

These sources provide the verified study basis while official Course 5043B detail is unavailable; they do not replace official RF hardware designs, authorised radar calibration, certified equipment specs or authorised strategic plans.